

Fig 7.11 *Resultant Force developed*

turn it about in a direction perpendicular to both the field and the current.

The fig 7.12a shows the upper side of the conductor, the magnetizing force of the field due to the current are additive while on the lower side these are subtractive. The result is to increase the flux density into the region directly above the conductor and to reduce flux density below the conductor.

If the current is reversed in the conductor, the strengthening of flux lines occur below the conductor and the conductor will be pushed upwards as shown in fig 7.12b.

As for the reasons above, one of the coil side 'A' will be forced to move downwards while 'B' moves upward. These two forces are equal in magnitude and opposite to each other. (7.12c). These twisting forces make the armature to rotate through the two ends, that are supported by bearings.

But practically, a DC machine will have multiple conductors and each conductor will be experiencing a force F=BI l newton. These forces collectively produce a driving torque which sets the armature rotating. The machine is then said to be motoring.

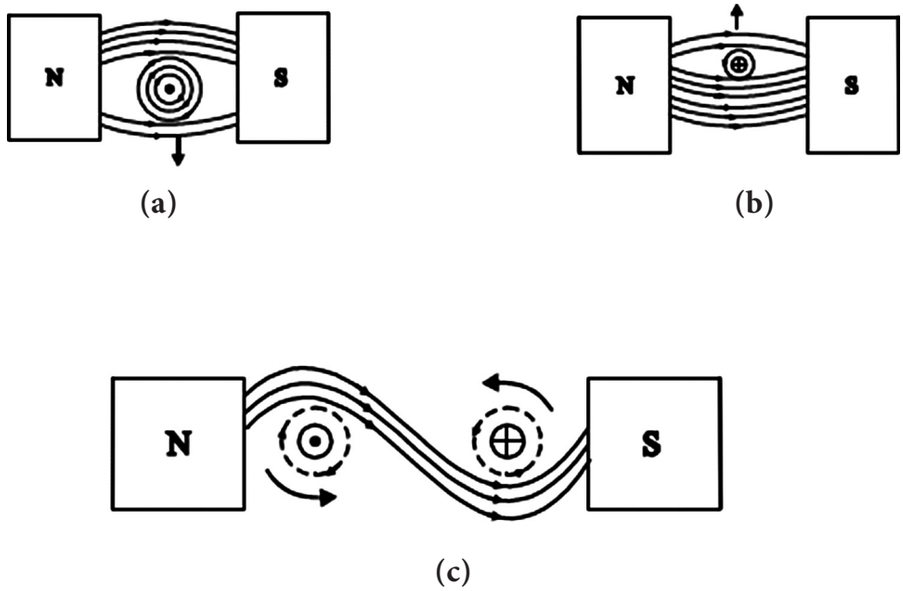


Fig 7.12 *Torque Development in DC Motors*

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